

VELAMMAL ENGINEERING COLLEGE

CHENNAI-600066

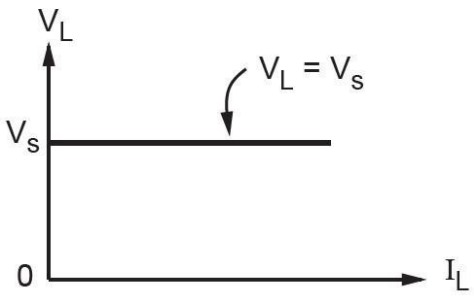
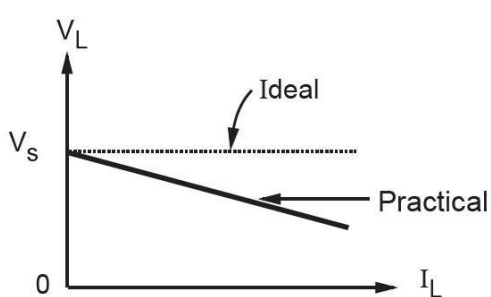
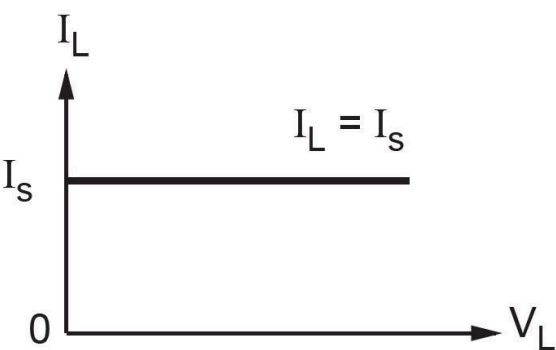
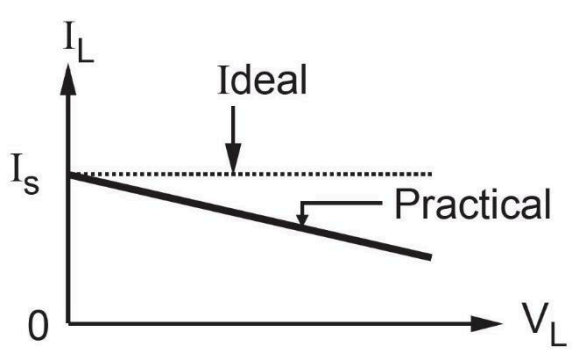
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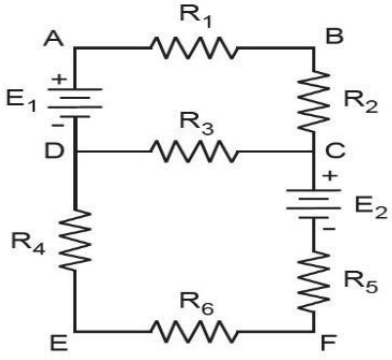
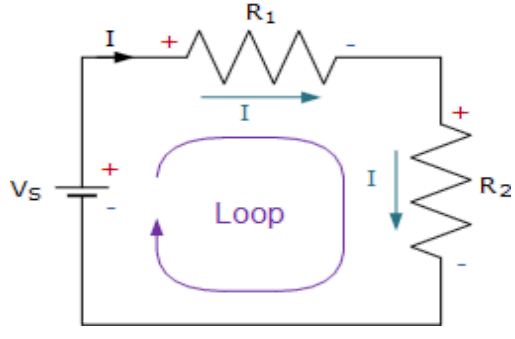
ELECTRONICS AND COMMUNICATION ENGINEERING

TWO MARKS WITH ANSWERS

23EC101T-ELECTRIC CIRCUIT AND ELECTRON DEVICES

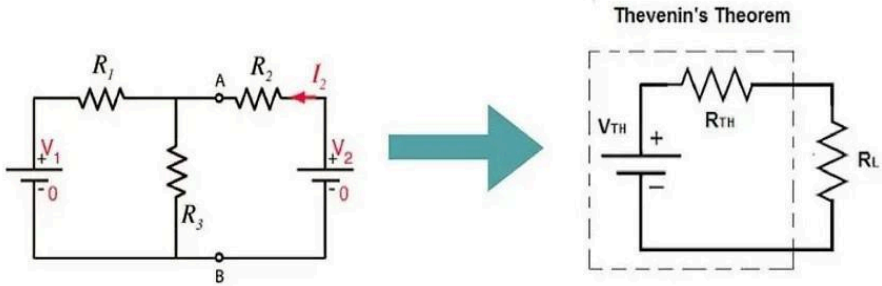
UNIT I - DC CIRCUIT ANALYSIS

1.	What is the difference between a circuit and a network? A network will not have any independent sources, whereas a circuit will have an independent source. When independent sources are connected to a network it becomes a circuit.
2.	Define Active and Passive Elements. The elements that can deliver energy are called Active elements. The elements which consume energy either by absorbing or storing it are called Passive elements.
3.	Define the Dependent source of a circuit. If the electrical energy supplied by a source depends on the voltage and current in some other part of the same circuit then it is called a Dependent source.
4.	 Ideal Voltage source.  Practical Voltage Sources
5.	 Ideal Current source.  Practical Current Sources
6.	Define Electrical Network An arrangement of various electrical energy sources along with the different circuit elements is called an electrical network
7.	Define Network Element

	Any individual circuit element with two terminals which can be connected to other circuit element is called a circuit Element.
8.	Define Junction point and Node of the Network A point where three or more branches meet is called a Junction point. The point D and C are Junction points shown in the below Electrical Network . A Point at which two or more elements are joined together i.e point at which two or more branches meet is called Node. The Junction point are also the nodes of the network. The points A,B,C,D,E,F are the nodes shown in the below Electrical Network. Electrical Network 
9.	Define a Loop of a network. Loop is a set of branches forming a closed path in a network in which if one branch is removed then remaining branch do not form a closed path. It is also defined as a closed path which originates from a particular node , terminating at the same node , travelling through various other nodes , without travelling through any node twice.  Loop of a Network
10.	How the Electrical Network are classified?

	Electrical circuits or networks <pre> graph TD A[Electrical circuits or networks] --> B1((1)) A --> B2((2)) A --> B3((3)) A --> B4((4)) B1 --> C1[Active] B1 --> C2[Passive] B2 --> C3[Linear] B2 --> C4[Nonlinear] B3 --> C5[Bilateral] B3 --> C6[Unilateral] B4 --> C7[Lumped] B4 --> C8[Distributed] </pre> Active Passive Linear Nonlinear Bilateral Unilateral Lumped Distributed
11.	Define Ohm’s Law It’s states that the current flowing through the circuit is directly proportional to the potential difference across the circuit and inversely proportional to the resistance of the circuit, provided the temperature remains constant. Mathematically $I = \frac{V}{R}$
12.	What are the limitations of Ohm’s Law? • It is not applicable to the non-linear devices such as diodes, Zener diodes, voltage regulators etc. • It does not hold good for non-metallic conductors such as silicon carbide.
13.	State Kirchoff’s Voltage Law It states that the “algebraic sum of Voltage drops across the circuit elements of any closed path is equal to the algebraic sum of emfs in the path”. In other words “ the algebraic sum of all the branch voltages, around any closed path or closed loop is always zero”. Around a closed path $\sum V = 0$
14.	State Kirchoff’s Current Law It states that “the total current flowing towards a junction point is equal to the total current flowing away from that junction point”. In other words “ the algebraic sum of all the current meeting at a junction point is always zero”. $\sum I \text{ at junction point} = 0$
15.	What is Short - Circuit? When any two points in a network are joined directly to each other with a conducting wire, the two points are said to be short-circuited. The resistance of such a circuit is Zero.

	The current I_{AB} is flowing through the circuited path. So Voltage across the cross circuit is always zero. According to Ohm's Law $V_{AB} = I_{AB} \times R_{AB} = 0$
16.	What is Open-Circuit? When there is no connection between the two points of a network, having some voltage across the two points, then the two points is said to be Open-circuited. As there is no direct connection in an open circuit, The resistance of such a circuit is infinite (∞). According to Ohms Law, $I_{AB} = \frac{V_{AB}}{R_{AB}} = \frac{V_{AB}}{\infty} = 0 \text{ A}$.
17.	Define Redundant Branch. In a circuit any branches or combinations of elements which do not carry any current then such branches or combinations are called redundant from circuit point of view. Redundant Branches
18.	State the Superposition theorem In any multisource complex network consisting of linear bilateral elements, the voltage across or current through, any given element of the network is equal to the sum of the individual voltages or currents, produced independently across or in that element by each source acting independently where all the remaining sources all replaced by their respective internal impedances.
19.	State Thevenin's theorem

	Any combination of linear bilateral circuit elements and active sources, connected to a given load R_L or Z_L replaced by a simple two terminal network consisting of Voltage source V_{th} volts and single Resistance (R_{eq}) or Impedance (Z_{eq}) in series with a voltage source across the two terminals of the load R_L or Z_L. V_{th} is Thevenin's equivalent voltage and R_{eq} is Equivalent Resistance. 
20.	State Norton's theorem Any combination of linear bilateral circuit elements and active sources, connected to a given load R_L or Z_L replaced by a simple two terminal network consisting of a single current source I_N and single Resistance (R_{eq}) or Impedance (Z_{eq}) in parallel with it across the two terminals of the load R_L or Z_L. I_N is the short circuit current flowing through the circuited path, replaced instead of R_L or Z_L. It is called as Norton's Current.
21.	State Maximum Power Transfer's theorem It states that maximum power transfer to the load, when the load resistance is equal to an equal resistance as viewed from the load terminals with load resistance removed and all the sources are replaced by the internal resistances.
22.	Write about the Network Reduction Technique Network reduction technique means to reduce the complicated network to a simple form consisting minimum number of elements to easily analyze the network. Two important Network reduction techniques are Source transformation and star-delta, delta-star transformation. The various theorems like the superposition theorem, Thevenin's theorem, and Norton's theorem also reduce the complicated network in to the simple network
23.	What is Supermesh? When a current source lies common to two meshes then the common current sources can be removed by analysis purpose and the resultant two meshes can be considered as one single mesh called Supermesh.
24.	What is Supernode? When a voltage source is connected between two nodes it can be short-circuited for analysis purposes and the short-circuited two nodes can be considered as one single node is called Supernode.

Unit-2 SINUSOIDAL STEADY STATE ANALYSIS

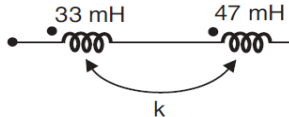
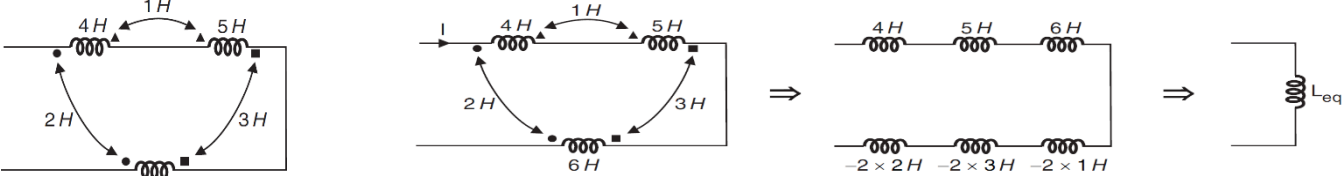
1.	A sinusoidal voltage is represented by the equation $100 \sin(503t + 30^\circ)$ volts. What is the frequency and time
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	period? Solution The general form of sinusoidal voltage is, $v(t) = V_m \sin(\omega t + \phi) \quad \dots\dots(1)$ Given that, $v(t) = 100 \sin(503t + 30^\circ) \quad \dots\dots(2)$ On comparing equations (1) and (2), we get, $\omega = 503 \text{ rad/s}$ Since, $\omega = 2\pi f$ $\text{Frequency, } f = \frac{\omega}{2\pi} = \frac{503}{2\pi} = 80 \text{ Hz}$ $\text{Time period, } T = \frac{1}{f} = \frac{1}{80} = 0.0125 \text{ second} = 12.5 \times 10^{-3} \text{ second} = 12.5 \text{ ms}$
2.	A sinusoidal current is given by the equation $i(t) = 7.072 \sin 314t$ A. What is the rms and average value of the current? <u>Solution</u> The general form of sinusoidal current is, $i(t) = I_m \sin(\omega t + \phi)$ Given that, $i(t) = 7.072 \sin 314t$ On comparing equations (1) and (2), we get, $I_m = 7.072 \text{ A}$ $\therefore \text{Rms value of current, } I = \frac{I_m}{\sqrt{2}} = \frac{7.072}{\sqrt{2}} = 5 \text{ A}$ $\text{Average value of current, } I_{\text{ave}} = \frac{2I_m}{\pi} = \frac{2 \times 7.072}{\pi} = 4.502 \text{ A}$
3.	Define form factor. Form factor is defined as the ratio of rms value and average value of a periodic waveform. $\text{Form factor, } k_f = \frac{\text{Average Value}}{\text{RMS Value}}$
4.	Define peak factor. Peak factor is defined as the ratio of peak value (or maximum value) and the rms value of a periodic waveform. $\text{Peak factor, } k_p = \frac{\text{RMS Value}}{\text{Maximum Value}}$
5.	What will be the inductance of a coil with 1000 turns while carrying a current of 2 A and producing a flux of 0.5 mWb? Given that, $N = 1000$, $\phi = 0.5 \text{ mWb}$, $I = 2 \text{ A}$

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6.	A steady current of 3 A flows through an inductance of 0.2 H. What will be the energy stored in the inductance? Given that, $I = 3A$, $L = 0.2H$ Energy stored in inductance, $W = \frac{1}{2}LI^2 = \frac{1}{2} \times 0.2 \times 3^2 = 0.9 \text{ Joules}$
7.	A 100 μF capacitance is charged to a steady voltage of 500 V. What will be the energy stored in the capacitance? Given that, $C = 100\mu F$, $V = 500V$ Energy stored in capacitance, $W = \frac{1}{2}CV^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 500^2 = 12.5 \text{ Joules}$
8.	<i>When a sinusoidal voltage $v = 200 \sin (377t + 30^\circ) V$ is applied to a load, it draws a current of $10 (\sin 377t + 60^\circ) A$. Determine the active and reactive power of the load.</i> The rms current and voltage phasors in polar form are, $\bar{V} = \frac{200}{\sqrt{2}} \angle 30^\circ V \quad ; \quad \bar{I} = \frac{10}{\sqrt{2}} \angle 60^\circ A$ Complex power, $\bar{S} = \bar{V} \bar{I}^* = \frac{200}{\sqrt{2}} \angle 30^\circ \times \left(\frac{10}{\sqrt{2}} \angle 60^\circ \right)^* = \frac{200}{\sqrt{2}} \angle 30^\circ \times \frac{10}{\sqrt{2}} \angle -60^\circ$$= \frac{2000}{2} \angle (30^\circ - 60^\circ) = 1000 \angle -30^\circ = 866 - j500 \text{ VA}$ Since, $\bar{S} = P + jQ$; Active power, $P = 866 W$ Reactive power, $Q = -500 \text{ VAR}$ (or 500 VAR-Capacitive)
9.	<i>A load consisting of 3Ω resistance and 4Ω inductive reactance draws a current of $10 A$ when connected to a sinusoidal source. Determine the voltage and power in the load.</i> Magnitude of impedance, $Z = \sqrt{R^2 + X_L^2} = \sqrt{3^2 + 4^2} = 5 \Omega$ Voltage, $V = IZ = 10 \times 5 = 50 V$ Power, $P = I^2 R = 10^2 \times 3 = 300 W$
10.	<i>When a sinusoidal voltage of $120 V$ is applied across a load, it draws a current of $8 A$ with a phase lead of 30°. Determine the resistance, reactance and impedance of the load.</i> Let, $\bar{V}$ be the reference phasor. $\therefore \bar{V} = 120 \angle 0^\circ V \quad \text{and} \quad \bar{I} = 8 \angle +30^\circ A$ Impedance, $\bar{Z} = \frac{\bar{V}}{\bar{I}} = \frac{120 \angle 0^\circ}{8 \angle 30^\circ} = 15 \angle -30^\circ = 12.99 - j7.5 \Omega$ Since, $\bar{Z} = R + jX$; Resistance, $R = 12.99 \Omega$ Reactance, $jX = -j7.5 \Omega$ (or 7.5Ω-Capacitive)

11.	What are coupled circuits? Coupled circuits refer to circuits involving elements with magnetic coupling. If the flux produced by an element of a circuit links other elements of the same circuit or a nearby circuit then the elements are said to have magnetic coupling.
12.	What are coupled coils? When two or more coils are linked by magnetic flux, the coils are called coupled coils (or coupled coils are group of two or more coils linked by magnetic flux).
13.	Define self-inductance. When permeability is constant, the self-inductance of a coil is defined as the ratio of flux linkage and current. (The flux linkage is the product of flux and number of turns.) $\therefore \text{ Self inductance, } L = \frac{N\phi}{i}$
14.	Define mutual inductance. When permeability is constant, the mutual inductance between two coupled coils is defined as the ratio of flux linkage in one coil due to common flux and current through another coil. (The flux linkage in one coil due to common flux is the product of the flux linking both the coils and the number of turns of the coil.) $\text{Mutual inductance, } M = \frac{N_1\phi_{21}}{i_2} \quad \text{or} \quad M = \frac{N_2\phi_{12}}{i_1}$ where, $\phi_{12} = \phi_{21}$ = Common flux. N_1, N_2 = Number of turns in coils 1 and 2. i_1, i_2 = Currents in coils 1 and 2.
15.	Define coefficient of coupling. In coupled coils, the coefficient of coupling is defined as the fraction of the total flux produced by one coil linking another coil. $\therefore \text{ Coefficient of coupling, } k = \frac{\phi_{12}}{\phi_1} \quad \text{or} \quad k = \frac{\phi_{21}}{\phi_2}$ where, ϕ_{12} = Flux produced by coil-1 linking coil-2. ϕ_{21} = Flux produced by coil-2 linking coil-1. ϕ_1 = Flux produced by coil-1. ϕ_2 = Flux produced by coil-2.
16.	Write the expression which relates self- and mutual inductances. Mutual inductance between two coils linked by magnetic flux is given by, Mutual inductance $M = k \sqrt{L_1 L_2}$ where, L_1 = Self-inductance of coil-1. L_2 = Self-inductance of coil-2. k = Coefficient of coupling.
17.	Write the expression for the equivalent inductance of two coupled coils connected in series. In series aiding, equivalent inductance, $L_{eq} = L_1 + L_2 + 2M$. In series opposing, equivalent reactance, $L_{eq} = L_1 + L_2 - 2M$.

18.	<i>Write the expression for the equivalent inductance of two coupled coils connected in parallel.</i> In parallel aiding, equivalent inductance, $L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$ In parallel opposing, equivalent inductance, $L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$
19.	<i>Determine the equivalent inductance of the circuit shown in Fig. Q3.56, if the coefficient of coupling (k) between the two coils is 0.6. (AU Dec'14, 2 Marks)</i> Mutual inductance, $M = k\sqrt{L_1 L_2} = 0.6 \times \sqrt{33 \times 47} = 23.6296 \text{ mH}$ Equivalent inductance, $L_{eq} = L_1 + L_2 + 2M$ $= 33 + 47 + 2 \times 23.6296 = 127.2592 \text{ mH}$  Fig. 3.56.
20.	Determine the Equivalent Inductance of series-connected inductances  Equivalent inductance, $L_{eq} = 4 + 5 + 6 - 2 \times 1 - 2 \times 3 - 2 \times 2 = 3H$

Unit-3 TRANSIENT ANALYSIS AND MAGNETIC COUPLED CIRCUITS

1.	What is meant by steady state of a network? A network in which branch current and node voltages are not changing with respect to time is said to be in steady state. Also, if the voltages and current in the circuit are having constant amplitude and frequency throughout the time interval of these parameter measurement, then such network is said to be in steady state.
2.	What is transient? The state (or condition) of a circuit from the instant of switching to the attainment of a steady state is called transient state or simply transient.
3.	Define time constant of an RL circuit The time constant of an RL circuit is defined as the time taken by the current through the inductance to reach its steady value if the initial rate of rise is maintained.
4.	<i>Find the time constant of an RL circuit with $R = 10\Omega$ and $L = 20 \text{ mH}$?</i> Time constant, $\tau = \frac{L}{R} = \frac{20 \times 10^{-3}}{10} = 2 \times 10^{-3} \text{ second} = 2 \text{ ms}$
5.	Define time constant of an RC circuit. The time constant of an RC circuit is defined as the time taken by the voltage across the capacitance to reach its steady value if the initial rate of rise is maintained.
6.	<i>What is the time constant of an RC circuit with $R = 10k\Omega$ and $C = 40\mu\text{F}$?</i> Time constant $\tau = RC = 10 \times 10^3 \times 40 \times 10^{-6} = 0.4 \text{ second}$

7. Why does transient occur in electric circuits?

The inductance does not allow a sudden change in current and the capacitance does not allow a sudden change in voltage. Hence, in inductive and capacitive circuits (or in general in reactive circuits), transient occurs during a switching operation.

8. What is the time constant of an RL circuit shown in Fig. Q4.9.1?

Let us find an equivalent inductance at terminals A-B using Thevenin's theorem as shown below:

Using Thevenin's equivalent, the given network can be drawn as shown in Fig. Q4.9.5.

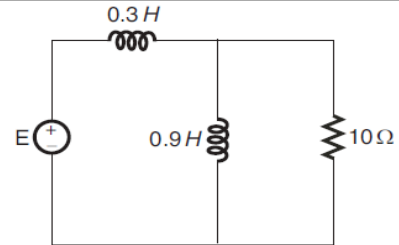
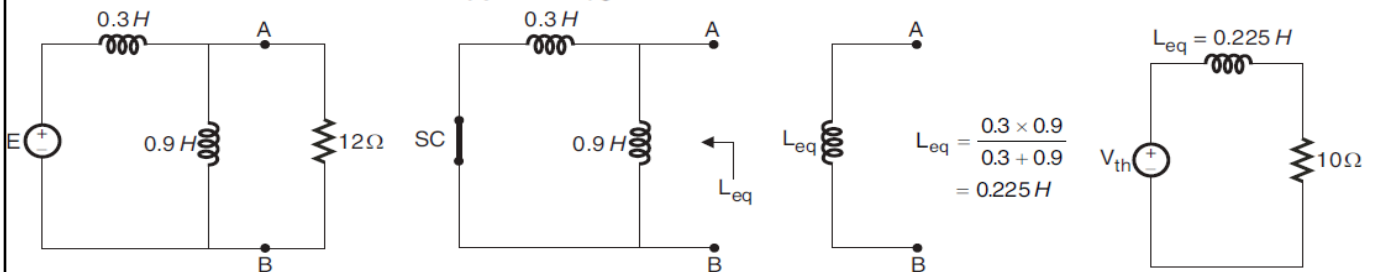


Fig. Q4.9.1.

Now, Time constant, $\tau = \frac{L_{eq}}{R} = \frac{0.225}{10} = 0.0225 \text{ second} = 22.5 \text{ ms}$



9. What is the time constant of an RC circuit shown in Fig. Q4.12.1?

Let us find an equivalent resistance at terminals A-B using Thevenin's theorem as shown below:

Using Thevenin's equivalent, the given network can be drawn as shown in Fig. Q4.12.5.

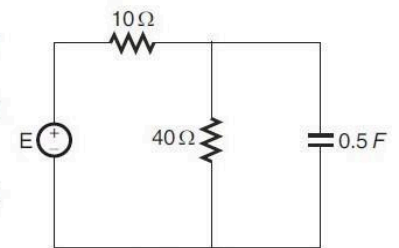
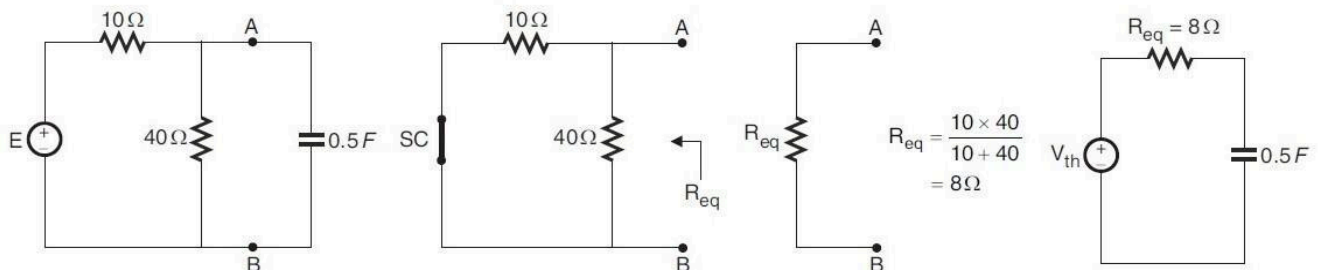


Fig. Q4.12.1.

Now, Time constant, $\tau = R_{eq} C = 8 \times 0.5 = 4 \text{ seconds.}$



10. An RL series circuit with $R = 10\Omega$ is excited by a dc voltage source of 30 V by closing the switch at $t = 0$. Determine the current in the circuit at $t = 2\tau$.

$$\begin{aligned} \text{Current, } i(t) &= \frac{E}{R} (1 - e^{-\frac{t}{\tau}}) \\ &= \frac{30}{10} (1 - e^{-2}) = 2.594 \text{ A} \end{aligned}$$

11.	<i>An RC series circuit is excited by a dc voltage source of 80 V by closing the switch at $t = 0$. Determine the voltage across the capacitor in a time of one time constant.</i> Voltage across capacitor, $v_C(t) = E(1 - e^{-\frac{t}{\tau}})$ $= 80(1 - e^{-1}) = 50.5696 \text{ V}$
12.	<i>A 50 μF capacitor is discharged through a 100 kΩ resistor. If the capacitor is initially charged to 400 V, determine the initial energy.</i> Initial energy $= \frac{1}{2} CV_0^2 = \frac{1}{2} \times 50 \times 10^{-6} \times 400^2$ $= 4 \text{ Joules}$
13.	
14.	What is damping ratio? The ratio of resistance of a circuit and resistance for critical damping is called damping ratio.
15.	What is critical damping? Critical damping is the condition of a circuit at which the oscillations in the response are just eliminated. This is possible by increasing the value of resistance in the circuit.
16.	What is natural and damped frequency? The response of a circuit is completely oscillatory with a frequency ω_n in the absence of resistance and frequency ω_n is called natural frequency. The response of an underdamped circuit is oscillatory with frequency of ω_d and these oscillations are damped as t tends to infinity. The frequency of damped oscillation response is called damped frequency.
17.	Write the condition for underdamping and critical damping in an RLC series circuit. The condition for underdamping is, $\left(\frac{R}{2L}\right)^2 < \frac{1}{LC}$ The condition for critical damping is, $\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$
18.	An RLC series circuit with $L = 2 \text{ H}$ and $C = 5 \text{ mF}$. Determine the value of R to give critical damping. Let, R_c = Value of R for critical damping $R_c = 2\sqrt{\frac{L}{C}} = 2 \times \sqrt{\frac{2}{5 \times 10^{-6}}} = 1264.911 \Omega$
19.	An RLC series circuit with $R = 10 \text{ W}$ and $L = 2 \text{ H}$. Determine the value of C to give critical

Unit 4 - DIODE CHARACTERISTICS

1.	Write down the diode current equation. The diode current equation relating the voltage V and current I is given by, $I = I_0 [e^{(V/\eta V_T)} - 1]$ I₀ – Reverse saturation current V – Applied Voltage η – 1 for germanium and 2 for silicon. V_T - Voltage equivalent of temperature in volts. V_T – kT Volts k – Boltzmann’s Constant (8.62 X 10⁻⁵ eV/°K) T – Temperature in °K										
2.	Differentiate between Zener breakdown and Avalanche breakdown. <table border="1" data-bbox="284 1098 1583 1533"> <thead> <tr> <th>zener breakdown</th><th>avalanche breakdown</th></tr> </thead> <tbody> <tr> <td>Breaking of covalent bond due to intense electric field.</td><td>Breaking of covalent bond due to collision.</td></tr> <tr> <td>This occurs when voltage is less than 6V.</td><td>This occurs when voltage is more than 6V.</td></tr> <tr> <td>The temperature coefficient is negative.</td><td>The temperature coefficient is positive.</td></tr> <tr> <td>The breakdown voltage decreases as junction temperature increases</td><td>The breakdown voltage increases as junction temperature increases.</td></tr> </tbody> </table>	zener breakdown	avalanche breakdown	Breaking of covalent bond due to intense electric field.	Breaking of covalent bond due to collision.	This occurs when voltage is less than 6V.	This occurs when voltage is more than 6V.	The temperature coefficient is negative.	The temperature coefficient is positive.	The breakdown voltage decreases as junction temperature increases	The breakdown voltage increases as junction temperature increases.
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3.	What is meant by barrier potential? The oppositely charged ions present on both sides of PN junction exert an electric potential across the junction even without any external voltage source applied. This potential is known as barrier potential.										
4.	List the diode parameters. The diode parameters are, ☐ Bulk Resistance ☐ Static Resistance										

	☐ Dynamic Resistance ☐ Reverse Resistance ☐ Knee voltage ☐ Break down voltage
5.	What is Avalanche breakdown? When reverse voltage is increased, the velocity of minority charge carriers increases. When such highly accelerated charge carrier collides against the atom in a covalent bond, it breaks the covalent bond and creates new charge carriers. Now these charge carriers gain sufficient velocity and further collides with the atoms and releases more charge carriers which leads to the breakdown of junction called avalanche breakdown or avalanche multiplication.
6.	Why silicon is widely used than germanium? The valance shell of silicon is in third shell while that of germanium is in fourth shell. Hence the valance electrons of germanium are far from the nucleus than silicon and therefore the valance electrons of germanium is loosely bound to nucleus than silicon. When a small energy is provided germanium electrons easily escape from the atom. Therefore at high temperature germanium is less stable than silicon and hence silicon is widely used than silicon.
7.	What is diffusion current in PN junction diode? When a semiconductor is non-uniformly doped, then there exists a concentration gradient. As a result the charge carriers move from higher concentration region to lower concentration region. This process is called diffusion and the current due to diffusion is known as diffusion current.
8.	What is meant by Zener breakdown? When a PN junction is heavily doped, the depletion region is very narrow. When a reverse voltage is applied to a diode, it causes a very intense electric field to appear across a narrow depletion region. Such an intense electric field is strong enough to pull some of the valence electrons into the conduction band by breaking their covalent bonds. These electrons then become free electrons which are available for conduction. A large number of such free electrons will constitute a large reverse current through the diode. This phenomenon is called zener breakdown.
9.	Define transition capacitance of a diode. A capacitance existing at the PN junction when the diode is reverse biased where the depletion regions act as plates while depletion region acts as dielectric is called transition capacitance of diode.

$$C_T = \frac{\epsilon A}{W}$$

	where ϵ = Permittivity of semiconductor $= \epsilon_0 \epsilon_r$ $\epsilon_0 = \frac{1}{36 \pi \times 10^9} = 8.849 \times 10^{-12} \text{ F/m}$ ϵ_r = Relative permittivity of semiconductor = 16 for Ge, 12 for Si A = Area of cross section and W = Width of depletion region
10.	Draw the VI characteristics of PN junction diode.
11.	Define storage time. When the diode is switched from forward bias to reverse bias, the minority charge carriers remain stored and decrease slowly to zero. This time for which minority charge carriers remain at zero is called storage time.
12.	Define mass action law. According to the law of mass action, the product of concentrations of holes and electrons is always constant at a fixed temperature. $np = n_i^2$ Where, n – Concentration of free electrons p - Concentration of holes

	n_i – Intrinsic concentration
13.	What is LCD? Liquid crystal display refers to the fact that these compounds have a crystalline arrangement of molecules, yet flow like a liquid. LCD's do not emit or generate light, rather alter externally generated illumination. A material in which crystalline arrangement of molecules is disturbed when electric field is applied; causing change in optical properties is called liquid crystal display.
14.	Define phototransistor. A transistor providing internal current multiplication when exposed to light is called phototransistor.
15.	What is a photovoltaic cell? Photovoltaic cell is nothing but a solar cell, which works on the principle of photovoltaic effect. Solar cell converts light energy into electrical energy, i.e.) voltage is proportional to the light incident on it.
16.	What is CCD? The charge coupled device is a part of a broader class of structure known as charge transfer device. These are dynamic devices which move charge along a pre-determined path under control of clock pulses.
17.	Mention some advantages and disadvantages of LCD. Advantages: - Less power consumption. - Uniform brightness with good contrast. - Low cost. - Low operating voltage and current. Disadvantages: - Limited temperature range. - Requires an ac drive.
18.	What is the principle of operation of LCD? When LCD cell is activated, molecular turbulence causes light to be scattered in all directions so that activated areas appear bright. This phenomenon is known as dynamic scattering.

19. Write down the significance of optocoupler.

A package which is a combination of light source like LED and light detector such as photodiode is called an opto coupler or opto isolator. In an opto coupler if input voltage changes then light emitted by light source changes. This changes the characteristics of a light detector and hence output voltage changes. Thus, the device couples input to output circuit optically providing electrical isolation.

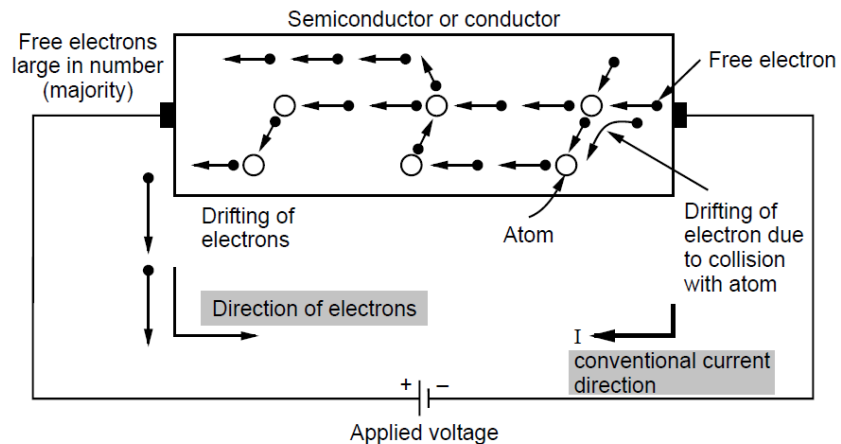
20. List the applications of LED.

- ☐ Used in all kinds of displays used in watches and calculators.
- ☐ In opto couplers.
- ☐ As ON – OFF indicator in various electronic circuits.
- ☐ In remote controls and burglar alarms.

21. Define Drift Current and Diffusion Current

Drift current:

When a voltage is applied to a material, the free electrons move towards the positive of battery. While moving they collide with the adjacent atom and keep changing their direction randomly. Free electrons still drifting towards the positive of the battery. This is called drift of charge carriers and the current due to such drifting of charge carriers is called **Drift current**.



Diffusion Current:

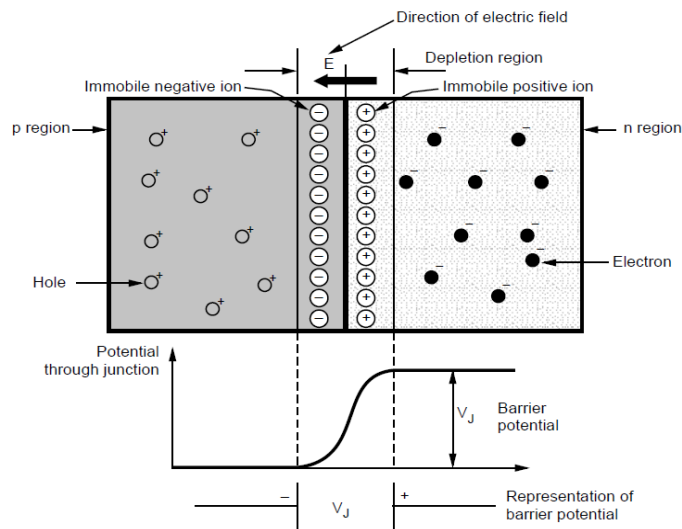
In this case, external voltage is not required. Due to non-uniform doping of the material a concentration gradient is created across the material due to which the charge carriers move from higher to lower concentration area. This is called diffusion and corresponding current is called **Diffusion Current**.

22	Define Diffusion capacitance of a diode. In forward biased condition, the width of the depletion region decreases, holes from p side get diffused in n side while electron from n side diffused in p side. As the applied voltage increases, concentration of injected charge particle increases. This rate of change of the injected charge with applied voltage is defined as the capacitance called diffusion capacitance. $C_D = \frac{dQ}{dV} = \frac{\tau I}{\eta V_T}$ τ – mean life time for holes
23	Define Knee voltage or a cut-in voltage When diode is forward is biased, some voltage is necessary to overcome barrier potential to make diode conduct. This is called its cut-in voltage. The minimum voltage at which the diode starts conducting and current starts increasing exponentially is called cut-in voltage, of forward voltage, break-point voltage, threshold voltage or knee voltage. It is denoted as V_γ and its value is 0.2 for germanium while 0.6 for silicon. Below this voltage, the diode current is very small and practically zero.

24

What is meant by depletion region

In a P-N junction, the diffusion of holes and electrons start initially. Near the junction, holes recombine in n region to form immobile positive ions. Similarly, electrons recombine in p-region to form immobile negative ions. With sufficient accumulation of such immobile ions on both sides, the diffusion stops. So, near the junction, there exists a region in which immobile positive and negative charges reside while mobile charge carriers in this region are completely depleted. This region is called depletion region.



25

State the expression for the dynamic resistance of a diode.

The dynamic resistance of a diode is given by

$$r = \frac{\eta V_T}{I_0 e^{\frac{V}{\eta V_T}}}$$

I_0 – Reverse saturation current

V – Applied Voltage

η – 1 for germanium and 2 for silicon.

V_T – Voltage equivalent of temperature in volts.

26

Compare the Silicon and Germanium diodes

Parameter	Silicon (Si)	Germanium (Ge)
Barrier Potential	0.7 V	0.3 V
Reverse Saturation Current	Few nA	Few μ A

23EC101T- ELECTRIC CIRCUIT AND ELECTRON DEVICES

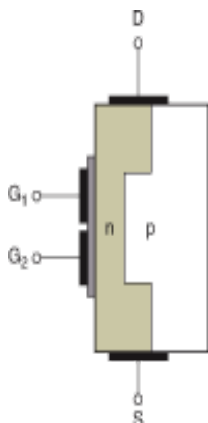
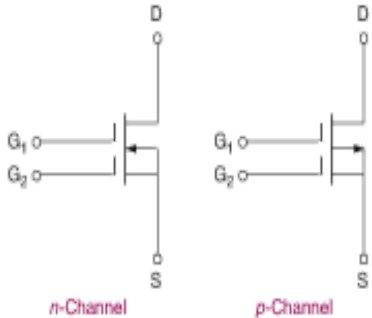
		Reverse Breakdown Voltage	Higher than Ge	Lower than Si
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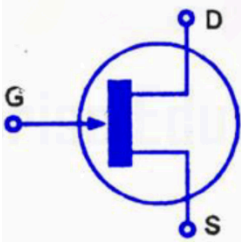
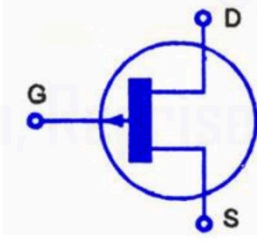
Unit 5 - TRANSISTOR CHARACTERISTICS

1.	What is FET? The field effect transistor (FET) is a semiconductor device which depends on the control of current by an electric field for its operation.
2.	Give the current voltage relationship of D-MOSFET and E-MOSFET. The current voltage relationship of D-Mosfet and E-Mosfet is given by,

	$I_D = I_{DSS} [1 - (\frac{V_{GS}}{V_P})]^2$ for D-MOSFET and $I_D = K[V_{GS} - V_T]^2$ for E-MOSFET. Where, I_D – Drain current. I_{DSS} – Drain to source current, when $V_{GS} = 0$. V_{GS} – Gate to source voltage. V_P – Pinch off voltage V_T – Threshold voltage K – Constant (Conduction parameter).
3.	List the features of FET. ☐ It is less noisy than a BJT. ☐ It has high input impedance. ☐ It has thermal stability ☐ It is relatively immune to radiation
4.	Why FET is called a unipolar device? In FET, the current is carried by only one type of charge carrier, either by electrons or holes. Hence it is known as unipolar device.
5.	Write the drain current equation or Shockley's equation of JFET? $I_D = I_{DSS} \{1 - (V_{GS}/V_P)\}^2$ Where, I_D – Drain current I_{DSS} – Saturation drain current V_{GS} – Gate to Source voltage V_P – pinch off voltage
6.	What are the characteristics associated with JFET? There are two important characteristics of JFET, ➤ Transfer characteristics – the graph drawn between gate source voltage V_{GS} and drain current I_D while keeping drain source voltage V_{DS} constant. ➤ Drain characteristics – drain characteristics shows the relationship between drain source voltage V_{DS} and drain current I_D.
7.	Define pinch-off voltage. With increase in the reverse bias of gate terminal, the depletion layer widens and a point is

	reached where entire channel is filled with depletion layer and drain current will be reduced to zero except for small reverse saturation currents order of nano amperes. The gate to source voltage at which the entire channel will be depleted of charge carrier is called pinch-off voltage.																				
8.	List out the operating regions of JFET. The various operating regions of JFET are, ☒ Ohmic region ☒ Saturation region ☒ Breakdown region ☐ Pinch-off or cut-off region.																				
9.	Compare JFET and BJT. <table><tr><th>S.No</th><th>JFET</th><th>BJT</th></tr><tr><td>1.</td><td>Unipolar device (current conduction is only due to one type of majority carrier either electron or hole).</td><td>Bipolar device (current condition, by both types of carriers, i.e., majority and minority-electrons and holes)</td></tr><tr><td>2.</td><td>The operation depends on the control of a junction depletion width under reverse bias.</td><td>The operation depends on the injection of minority carriers across a forward biased junction.</td></tr><tr><td>3.</td><td>Voltage driven device. The current through the two terminals is controlled by a voltage at the third terminal (gate).</td><td>Current driven device. The current through the two terminals is controlled by a current at the third terminal (base).</td></tr><tr><td>4.</td><td>Low noise level.</td><td>High noise level.</td></tr><tr><td>5.</td><td>High input impedance(due to reverse bias).</td><td>Low input impedance (due to forward bias).</td></tr></table>			S.No	JFET	BJT	1.	Unipolar device (current conduction is only due to one type of majority carrier either electron or hole).	Bipolar device (current condition, by both types of carriers, i.e., majority and minority-electrons and holes)	2.	The operation depends on the control of a junction depletion width under reverse bias.	The operation depends on the injection of minority carriers across a forward biased junction.	3.	Voltage driven device. The current through the two terminals is controlled by a voltage at the third terminal (gate).	Current driven device. The current through the two terminals is controlled by a current at the third terminal (base).	4.	Low noise level.	High noise level.	5.	High input impedance(due to reverse bias).	Low input impedance (due to forward bias).
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	1.	JFET Gate is not insulated from the channel.	MOSFET Gate is insulated from the channel.
	2.	Can be operated in both depletion and enhancement mode.	Channel and gate forms parallel plate capacitor.
	3.	Can be operated in depletion mode only.	Can be operated in both depletion and enhancement mode.
	4.	Input impedance is high.	Input impedance is very high.
11.	How can a FET be used as a voltage controlled resistor? (or) In which region JFET acts as a resistor and why? In ohmic region, JFET acts as a resistor. The depletion layer of the channel is very small and JFET acts like a variable resistor. As V_{DS} is increased from zero, I_D corresponding to it increases. In ohmic region, drain current I_D varies with V_{DS} and drain to source resistance varies in linear manner. Thus FET acts as a voltage controlled resistor in ohmic region.		
12.	Define amplification factor in JFET. Amplification factor is defined as the ratio of change in drain to source voltage to Change in gate to source voltage when drain current is maintained constant. $\mu = \frac{\text{Change in } V_{DS}}{\text{Change in } V_{GS}}$		
13.	Draw the symbol for dual gate MOSFET.  (a) Construction  (b) Schematic symbols		
14.	What is transconductance in JFET? The ratio of change in drain current, ΔI_D , to the change in gate-source voltage, ΔV_{GS} , is the transconductance, gm. The unit of transconductance is the Siemens (S). It is the reciprocal of		

	resistance (Ω). $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$										
15.	What are the applications of JFET? The applications of JFET are, ☐ Used in RF amplifiers, FM turners ☐ Used as buffer amplifiers ☐ Used in oscillator circuits ☐ Used as buffer in measuring instruments.										
16.	What is channel length modulation in MOSFET? Channel length modulation in MOSFET is a phenomenon where, • An increase in V_{DS} causes the depletions region at the drain junction to grow. and • The length of the effective channel is reduced										
17.	Why input impedance of MOSFET is high when compared to JFET? There is no direct electrical connection between gate terminal and the channel of MOSFET. Gate is insulated from the channel by a thin layer of dielectric SiO_2. Due to this extra layer, input impedance of MOSFET is very high.										
18.	Compare N Channel JFET and P Channel JFET <table border="1"> <thead> <tr> <th>N Channel FET</th><th>P Channel FET</th></tr> </thead> <tbody> <tr> <td>1. Current carriers are Electrons</td><td>1. Current carriers are holes</td></tr> <tr> <td>2. Mobility of electrons is almost twice that of Holes in P channel FET</td><td>2. Mobility of holes is poor</td></tr> <tr> <td>3. Low input Noise</td><td>3. More noise</td></tr> <tr> <td>4. Large Transconductance</td><td>4. Low Transconductance</td></tr> </tbody> </table>	N Channel FET	P Channel FET	1. Current carriers are Electrons	1. Current carriers are holes	2. Mobility of electrons is almost twice that of Holes in P channel FET	2. Mobility of holes is poor	3. Low input Noise	3. More noise	4. Large Transconductance	4. Low Transconductance
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19.	Draw the symbol of n channel JFET and p channel JFET  n - Channel JFET  p - Channel JFET										

20.	
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